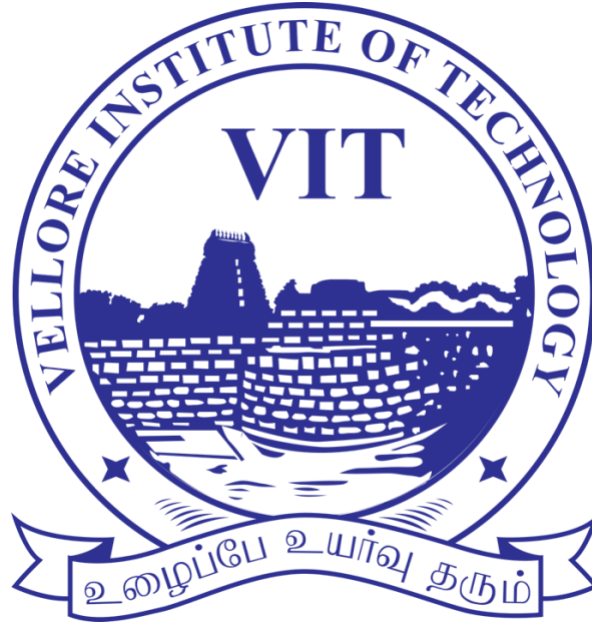




PROJECT TITLE: IDENTIFYING FAKE REVIEWS FOR PRODUCTS ON E-COMMERCE PLATFORMS

TEAM MEMBERS

Varsha S – 22BCE2469
Srija Puvvada – 22BCE3851
Shane Thomas – 22BCE3649
Apurba Koirala – 22BCE3799
Swikriti Shahi – 22BCE3869

SUBMITTED TO

Dr. Deepa K
Assistant Professor Sr. Grade 1
School of Computer Science and Engineering

UNDER

Foundations of Data Science – Course Project (BCSE206L)

Abstract

Fake online reviews have become a major concern for e-commerce platforms, as they manipulate product reputations, mislead consumers, and distort market competition. These deceptive reviews are intentionally crafted to appear genuine, making manual detection challenging and costly.

Fake review detection generally involves extracting textual and behavioral features, applying NLP techniques, and training models ranging from SVMs and Random Forests to LSTMs and transformers using inputs like TF-IDF, sentiment scores, n-grams, and posting patterns to classify reviews as genuine or fraudulent.

In this project, we propose a hybrid detection methodology that first trains a transformer-based deep learning model on a benchmark fake review dataset and then tests the model on real-world data obtained through automated web scraping from e-commerce platforms. The pre-processing pipeline will include text cleaning (removing HTML tags, emojis, and URLs), lowercasing, punctuation and stopword removal, lemmatization, tokenization, and TF-IDF/embedding generation. For web-scraped reviews, additional metadata such as reviewer ID, posting date, and rating deviation will be integrated to enhance detection accuracy.

While many existing methods rely mainly on textual features or synthetic datasets, our approach combines transformer-based semantic embeddings with real-world behavioral data to improve detection across varied and noisy environments. Testing the model on live scraped reviews helps ensure it performs well not only on benchmark datasets but also in practical, real-world applications.

Datasets:

Deceptive Opinion Spam dataset (Ott et al.)

Amazon GPT-2 fake review corpus

Yelp filtered reviews dataset